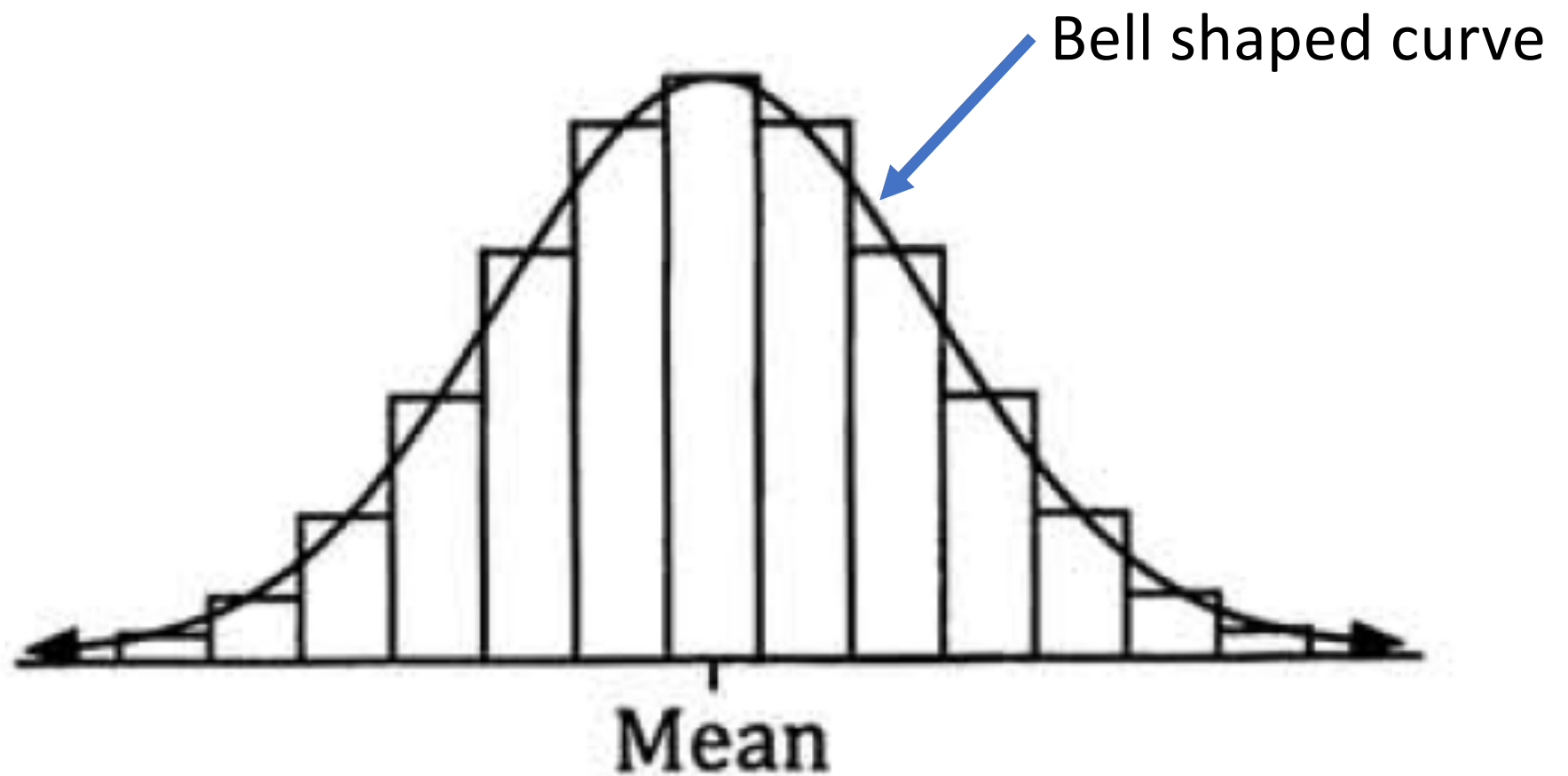


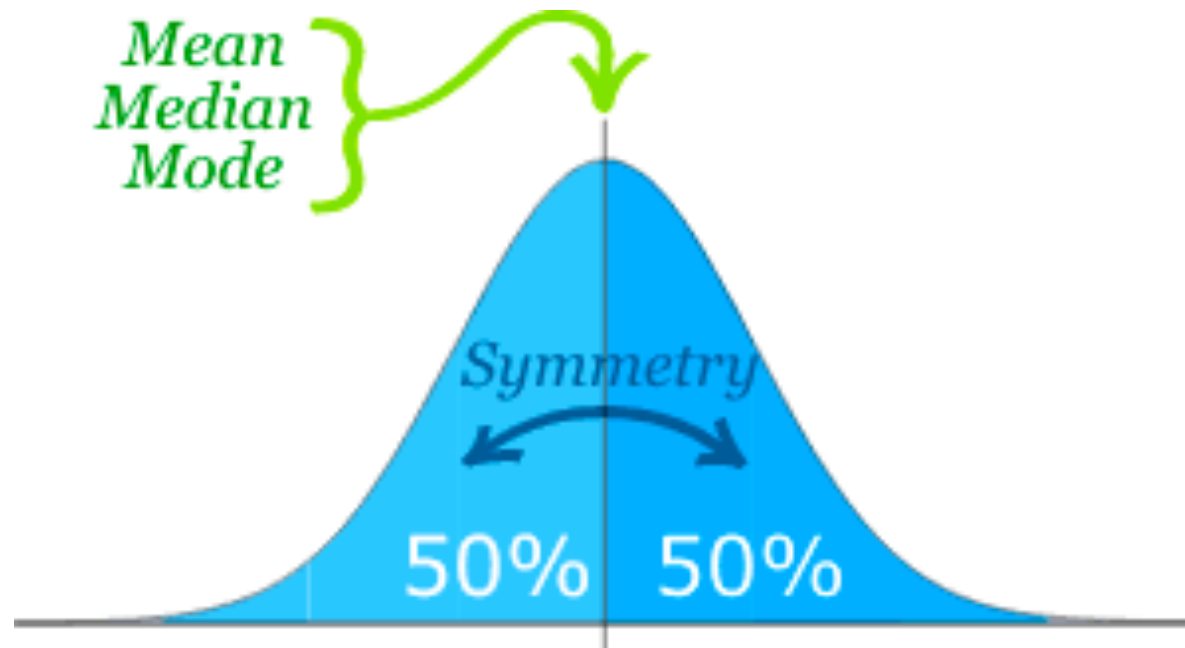
Normal Distribution

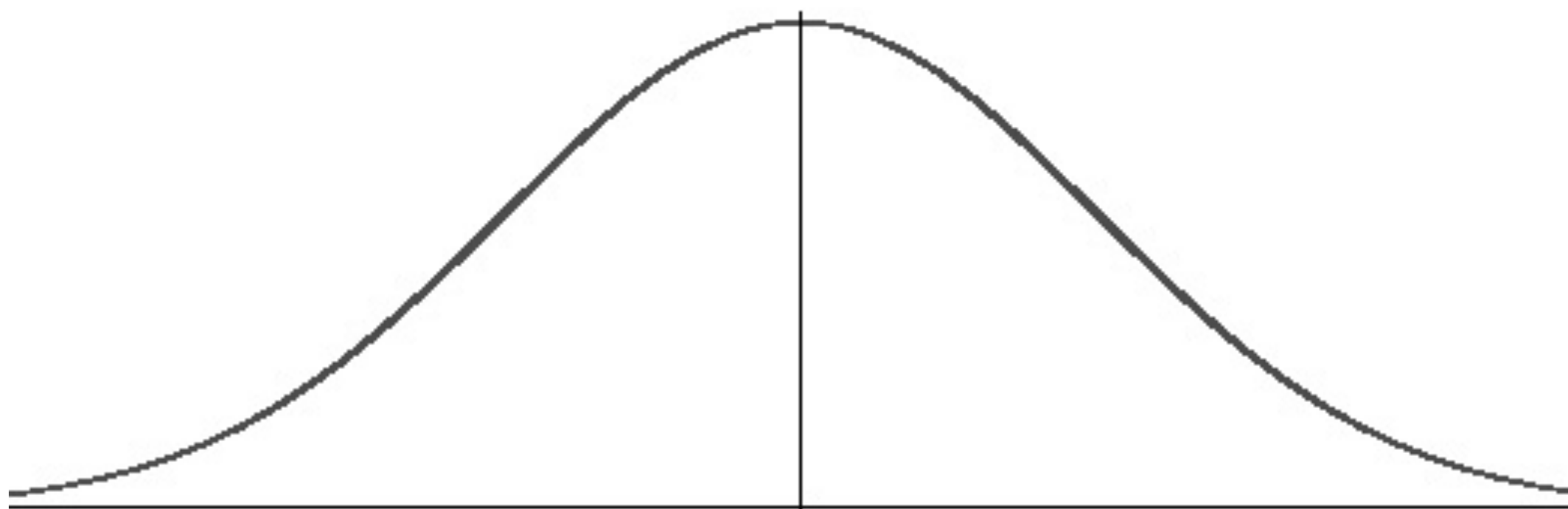


Normal Distribution

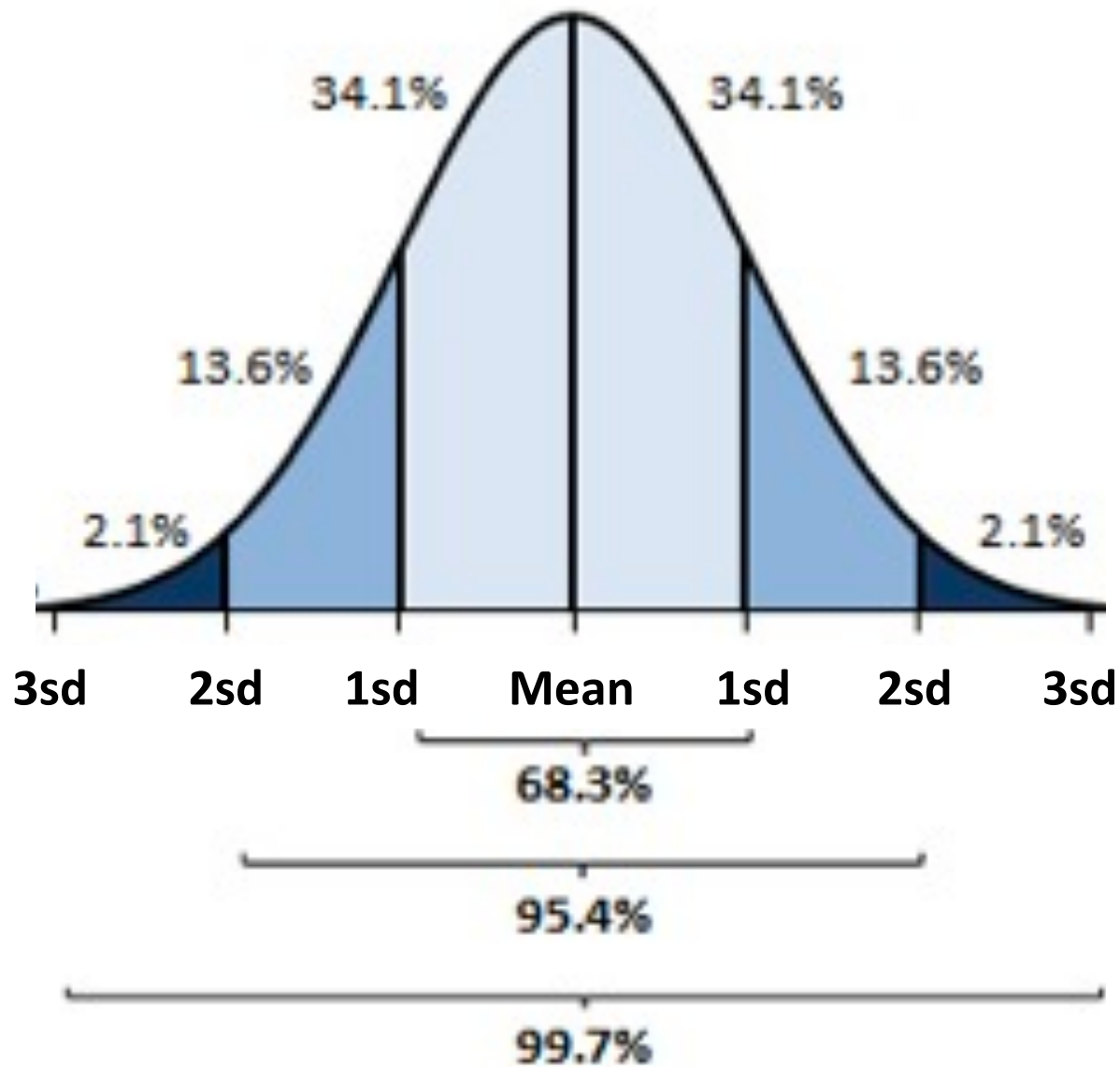
The normal distribution

- Also known as Gaussian distribution
- mean = median = mode
- symmetry about the centre
- 50% of values less than the mean and 50% greater than the mean





68%, 95%, 99.7% RULE

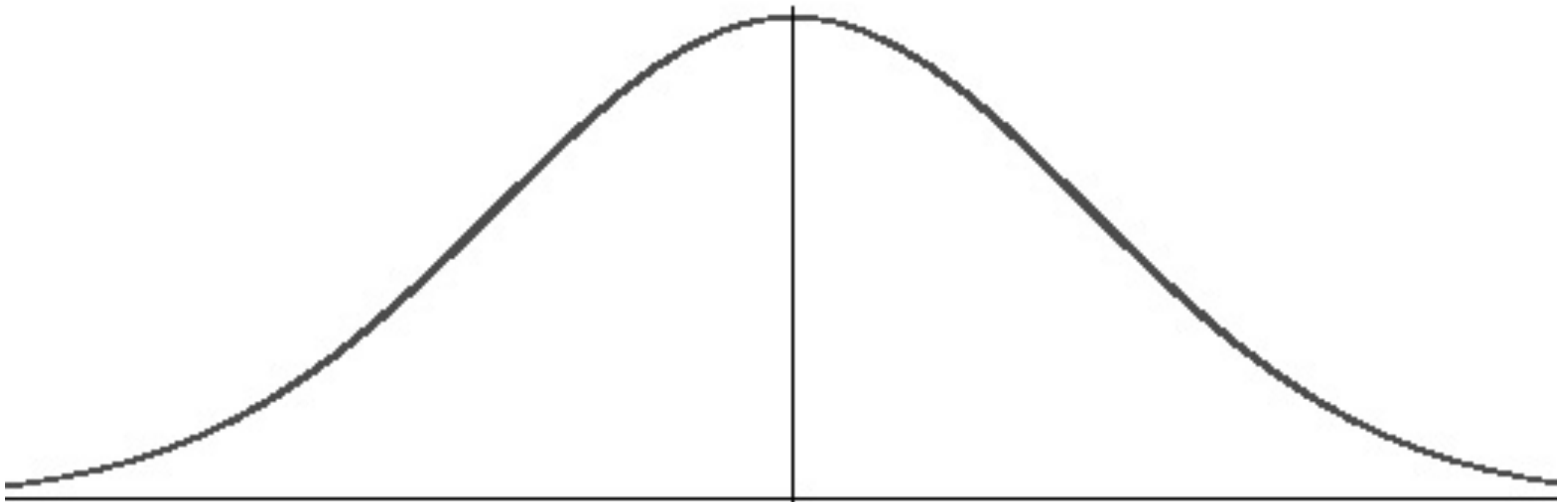


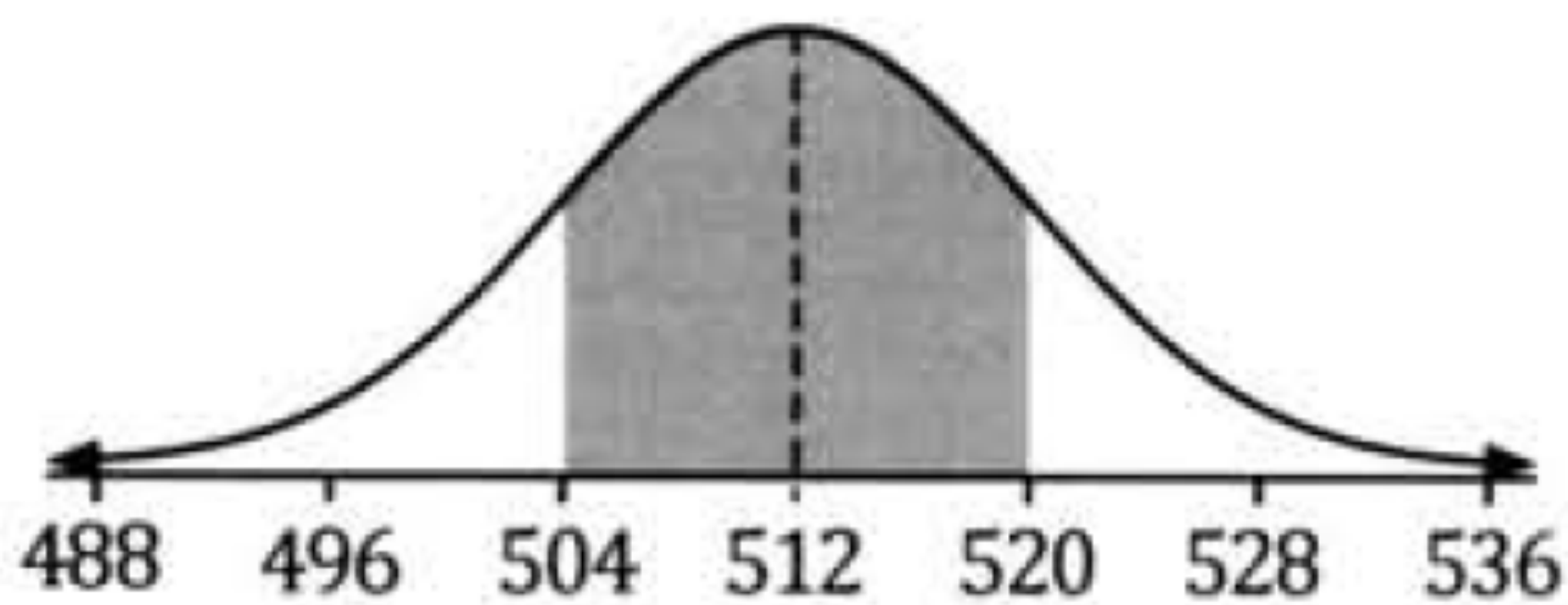
What does it mean?

- 68.3% of data values are within 1 standard deviation of the mean
 - Probability of a randomly chosen data point within $1\sigma = 68.3\%$
- 95.4% of data values are within 2 standard deviations of the mean
 - Probability of a randomly chosen data point within $2\sigma = 95.4\%$
- 99.7% of data values are within 3 standard deviations of the mean
 - Probability of a randomly chosen data point within $3\sigma = 99.7\%$

Example 1

A box of breakfast cereal has "contains 500 grams of breakfast cereal" printed on it. Suppose that in fact the weight of breakfast cereal contained in these boxes is normally distributed with a mean of 512 grams and a standard deviation of 8 grams. Determine the probability that a randomly chosen box of this cereal contains between 504 grams and 520 grams.

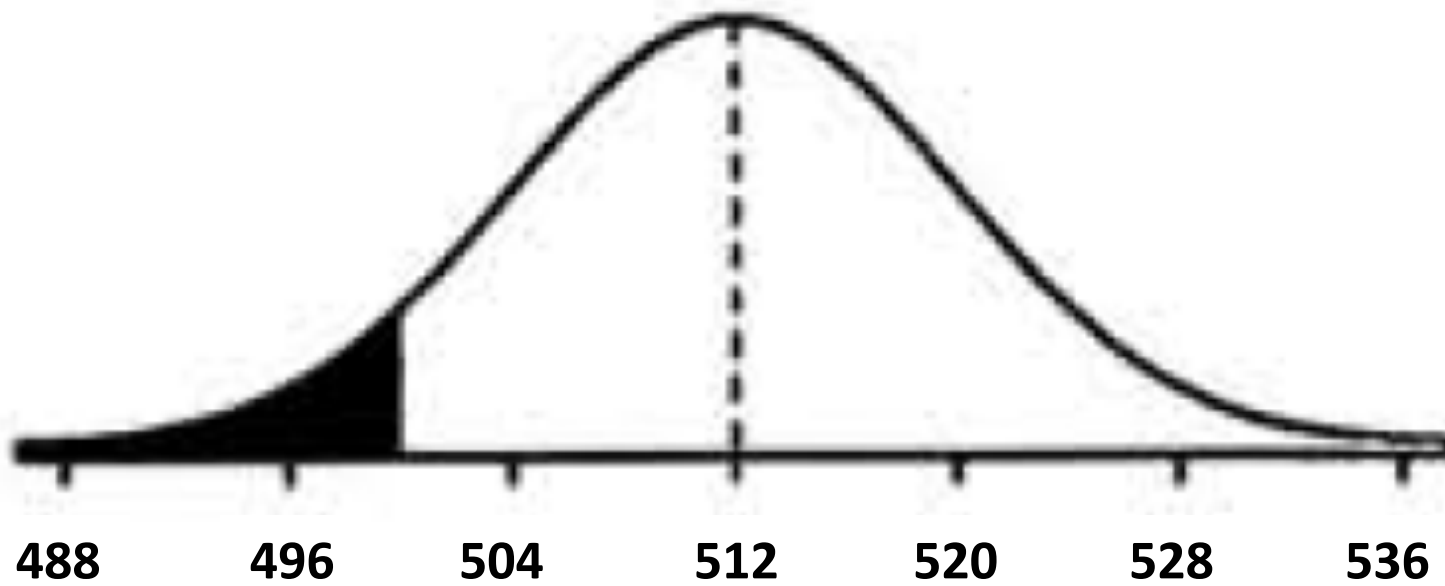


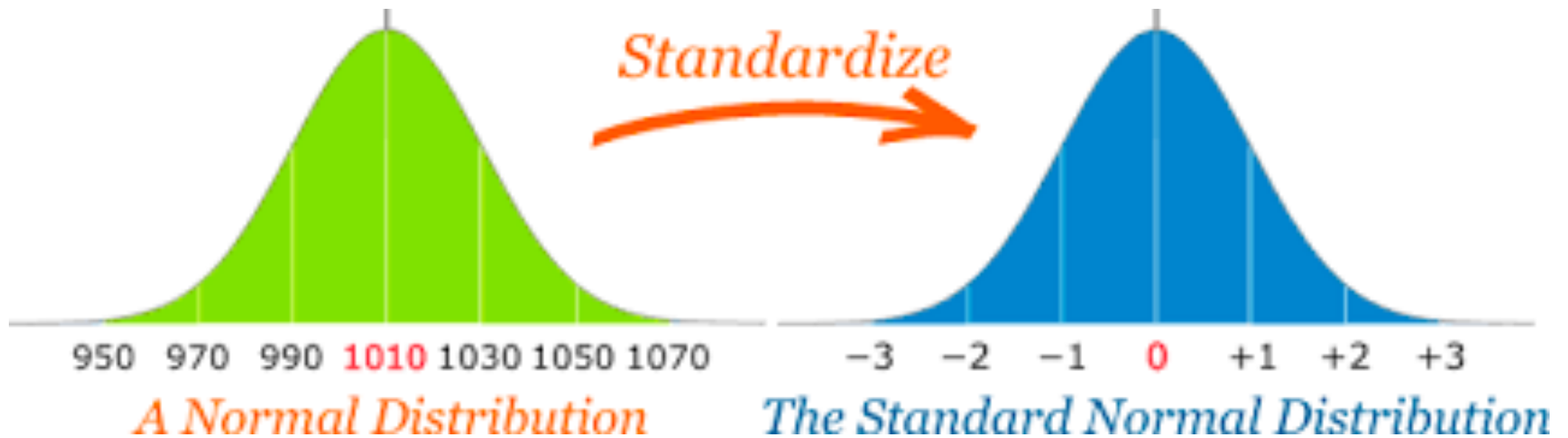


Example 2

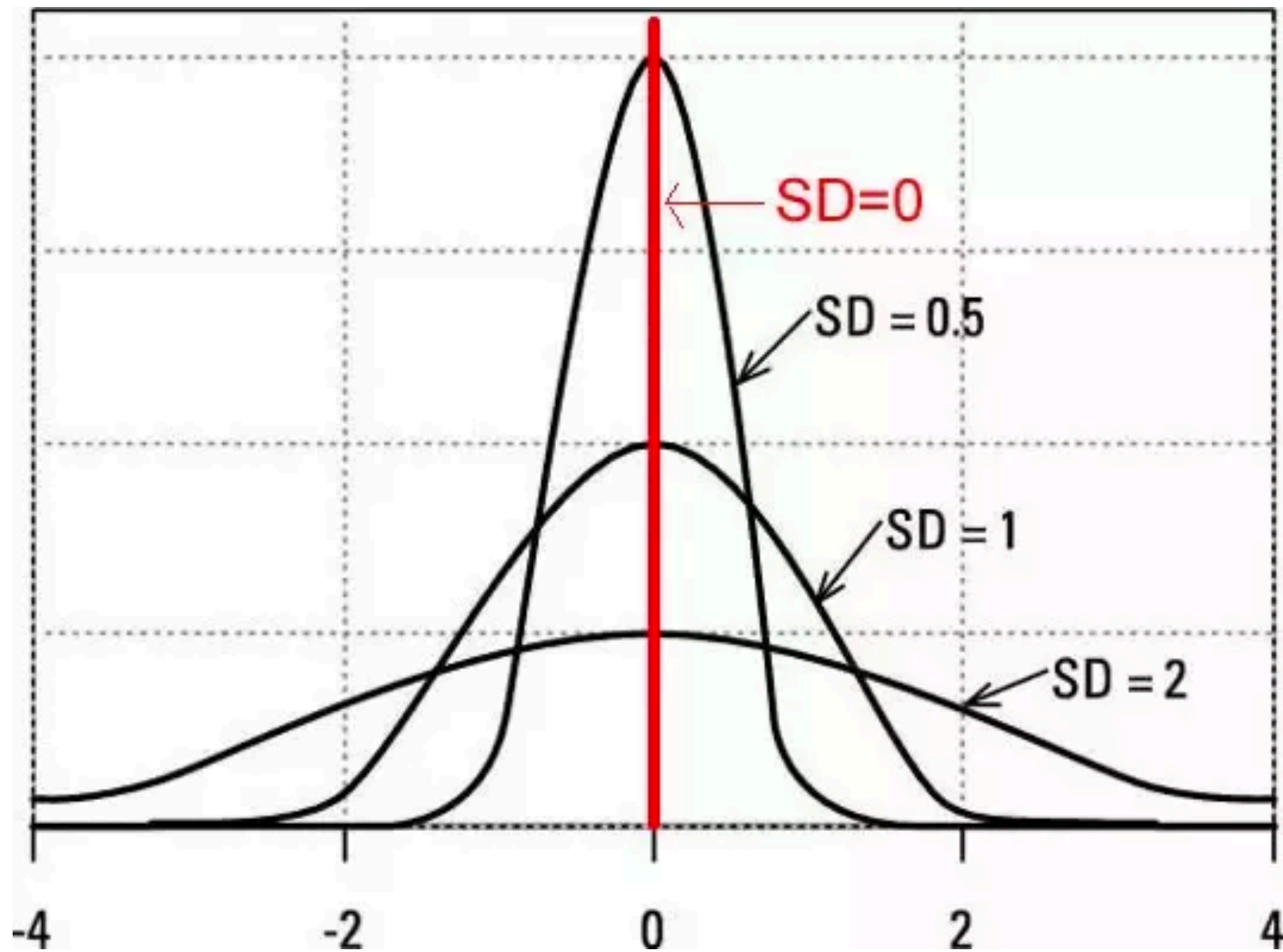
A box of breakfast cereal has "contains 500 grams of breakfast cereal" printed on it. Suppose that in fact the weight of breakfast cereal contained in these boxes is normally distributed with a mean of 512 grams and a standard deviation of 8 grams.

- (a) Determine the probability that a randomly chosen box of this cereal contains less than 500 grams.
- (b) In a random sample of 100 boxes of this cereal approximately how many boxes should we would expect to contain less than 500 g?





$$\text{Standardised score} = \frac{\text{raw score} - \text{mean}}{\text{standard deviation}}$$

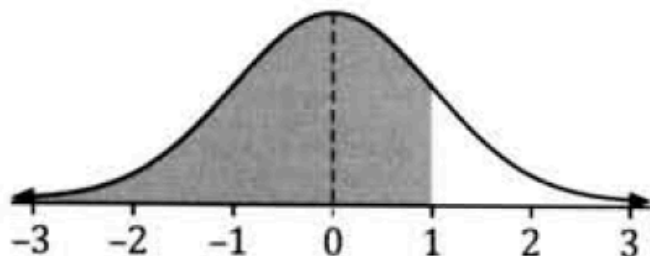


Notation

$$X \sim N(\mu, \sigma^2)$$

- X - normally distributed set of data
- μ - mean of the data
- σ - standard deviation (note: σ^2 = variance)

mean = 0
standard deviation = 1



$$P(X < 1) = 0.8413$$

Can you also get 0.84 using the
68%, 95%, 99.7% rule?

ClassPad Manager - default-2.vcp

CASIO

Lower $-\infty$
Upper 1
 σ 1
 μ 0

<< Back Help Next >>

Math1	Line	$\frac{\Box}{\Box}$	$\sqrt{\Box}$	π	$\Rightarrow$
Math2	Define	f	g	i	∞
Math3	solve(	dSlv	'	{ $\frac{\Box}{\Box}$ }	
Trig	<	>	()	{ }	[]
Var	$\leq$	$\geq$	=	$\neq$	<
abc	$\leftarrow$	$\rightarrow$	$\rightarrow$	ans	EXE

NormCD

Settings Menu Main Resize Swap Rotate Esc

Keyboard Shift Clear

= x y z ^ ÷

(7 8 9 ×

) 4 5 6 -

, 1 2 3 +

(-) 0 . EXP EXE

ClassPad Manager - default-2.vcp

CASIO

prob 0.8413447
z Low $-\infty$
z Up 1
 σ 1
 μ 0

<< Back Help

Math1	Line	$\frac{\Box}{\Box}$	$\sqrt{\Box}$	π	$\Rightarrow$
Math2	Define	f	g	i	∞
Math3	solve(	dSlv	'	{ $\frac{\Box}{\Box}$ }	
Trig	<	>	()	{ }	[]
Var	$\leq$	$\geq$	=	$\neq$	<
abc	$\leftarrow$	$\rightarrow$	$\rightarrow$	ans	EXE

NormCD

Settings Menu Main Resize Swap Rotate Esc

Keyboard Shift Clear

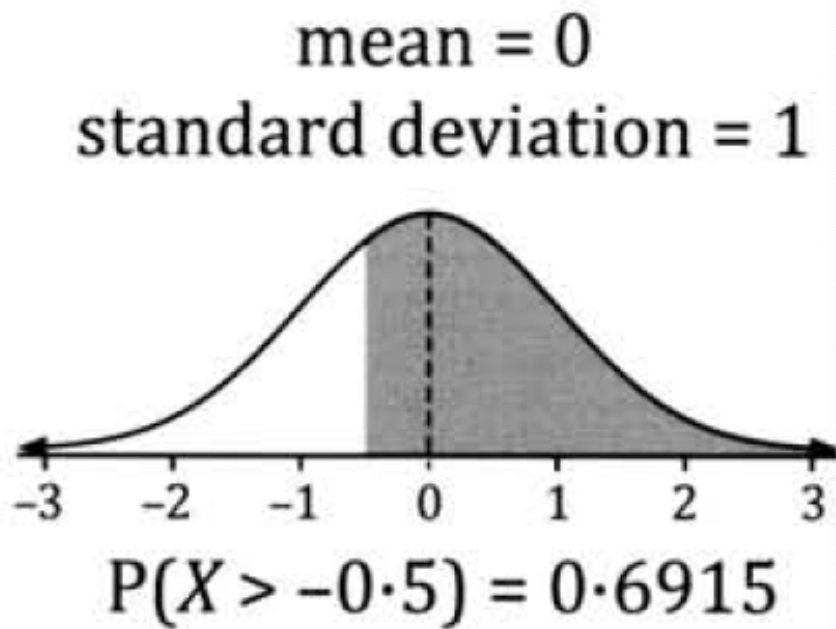
= x y z ^ ÷

(7 8 9 ×

) 4 5 6 -

, 1 2 3 +

(-) 0 . EXP EXE



ClassPad Manager - default-2.vcp

CASIO

NormCD

Lower
Upper
 σ
 μ

<< Back Help Next >>

Math1	Line		$\sqrt{\square}$	π	$\Rightarrow$
Math2	Define	f	g	i	∞
Math3	solve(	dSlv	'	$\left\{ \begin{smallmatrix} \square \\ \square \end{smallmatrix} \right\}$	
Trig	<	>	()	{ }	[]
Var	$\leq$	$\geq$	=	$\neq$	$\angle$
abc	$\leftarrow$			ans	EXE

Settings Menu Main Resize Swap Rotate Esc

Keyboard Shift Clear

= x y z ^ ÷

(7 8 9 ×

) 4 5 6 −

, 1 2 3 +

(−) 0 . EXP EXE

ClassPad Manager - default-2.vcp

CASIO

NormCD

prob
z Low
z Up
 σ
 μ

<< Back Help

Math1	Line		$\sqrt{\square}$	π	$\Rightarrow$
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Trig	<	>	()	{ }	[]
Var	$\leq$	$\geq$	=	$\neq$	$\angle$
abc	$\leftarrow$			ans	EXE

Settings Menu Main Resize Swap Rotate Esc

Keyboard Shift Clear

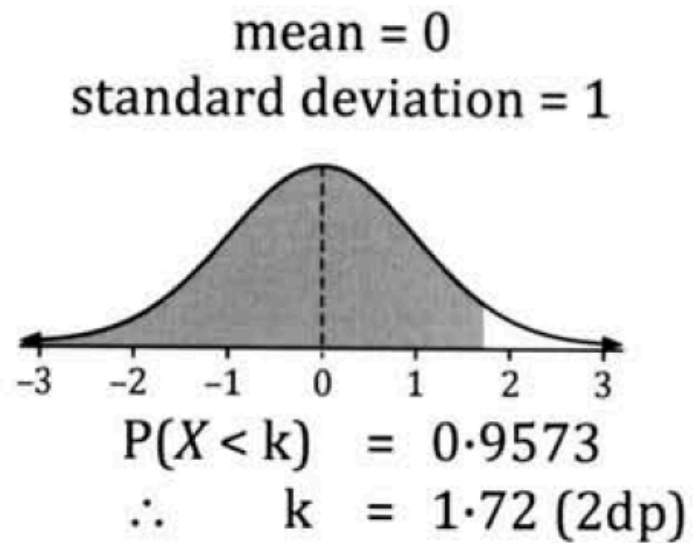
= x y z ^ ÷

(7 8 9 ×

) 4 5 6 −

, 1 2 3 +

(−) 0 . EXP EXE



ClassPad Manager - default-2.vcp

CASIO

	list 1	list 2	list 3
1			
2			
3			
4			
5			
6			

Cal▶

[1] =

Tail setting **Left**

prob 0.9573

σ 1

μ 0

<< Back Help Next >>

InvNormCD

Settings Menu Main Resize Swap Rotate Esc

Keyboard

Shift

Clear

= x y z $\wedge$ $\div$

(7 8 9 $\times$

) 4 5 6 $-$

, 1 2 3 $+$

(-) 0 $\cdot$ EXP EXE

ClassPad Manager - default-2.vcp

CASIO

	list 1	list 2	list 3
1			
2			
3			
4			
5			
6			

Cal▶

[1] =

x_1 InvN 1.7201785

prob 0.9573

σ 1

μ 0

<< Back Help

InvNormCD

Settings Menu Main Resize Swap Rotate Esc

Keyboard

Shift

Clear

= x y z $\wedge$ $\div$

(7 8 9 $\times$

) 4 5 6 $-$

, 1 2 3 $+$

(-) 0 $\cdot$ EXP EXE

Type Inv. Distribution

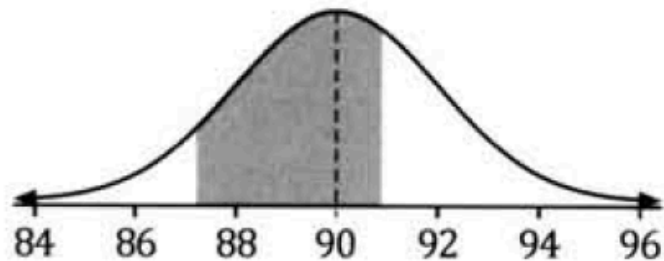
Inverse Normal CD

Help Next >>

Settings Menu Main Resize Swap Rotate Esc

Keyboard

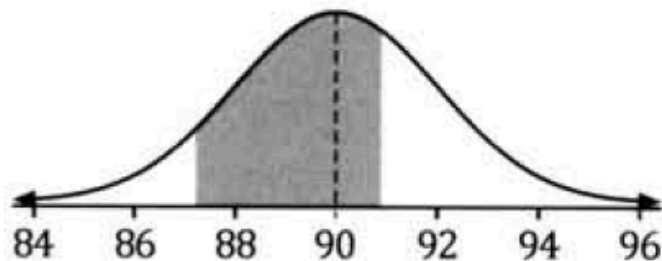
mean = 90
standard deviation = 2



$$P(87.2 < X < k) = 0.5964$$

$$\therefore k = 90.92 \text{ (2dp)}$$

Step 1: Find the $P(X > 87.2)$
using your class pad



prob	0.0807567
z Low	-5E+998
z Up	-1.4
σ	2
μ	90

Step 2: Add the two
probabilities together to
determine $P(X < K)$
 $0.5964 + 0.0808 = 0.6772$

Step 3: Use Inverse
Distribution to find K

